Hospital Operations & Financial Overview

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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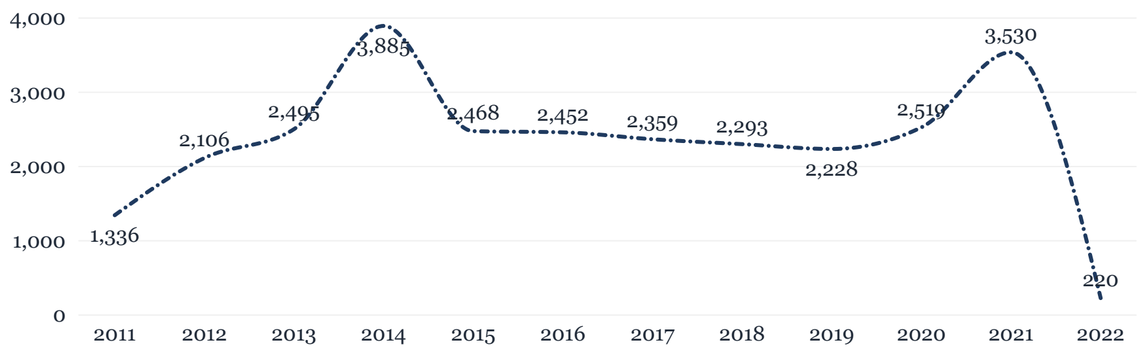
All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

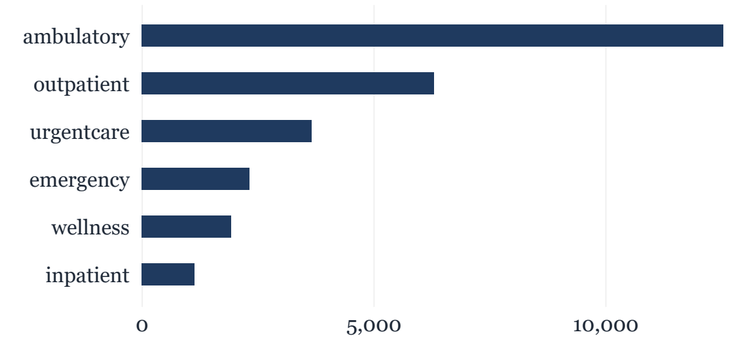
Prepared by Kukoyi Zainab Omotola

Total Encounters	Total Claim Cost	Zero Coverage %	Payer Coverage %	Avg Claim Cost per Encounter
27,891	$101.5M	48.7%	30.6%	$3,640

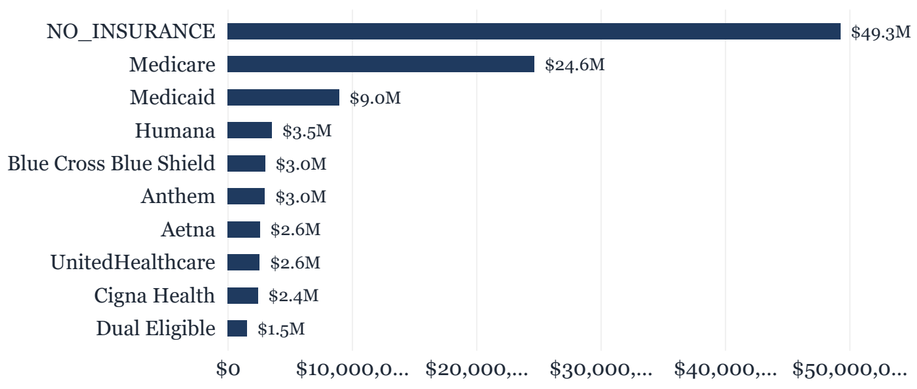
Encounters by Year



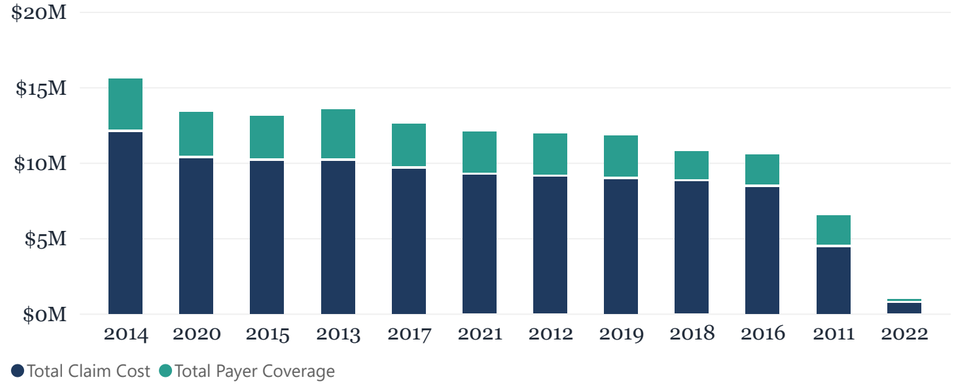
Encounters by Class



Claim Cost by Payers



Claim Cost vs Total Payer Coverage by Year



Footnote;2022 is partial(date ends 5 Feb 2022). Zero payer coverage may mean uninsured, unrecorded or unpaid

Hospital Operations Encounter Volume Trend

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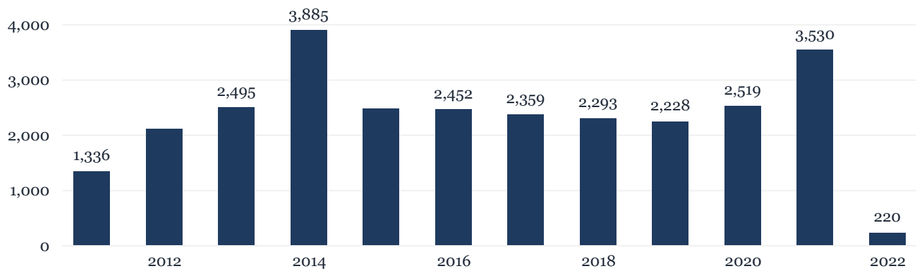
Prepared by Kukoyi Zainab Omotola

Total Encounters
27,891

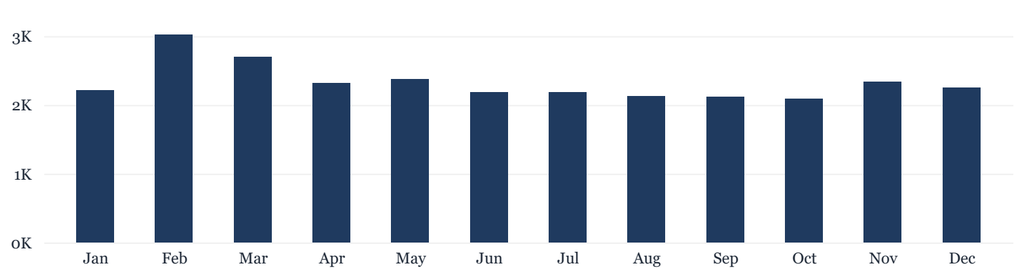
Unique Patients
974

Peak month: Feb 2014 (1,179 encounters)

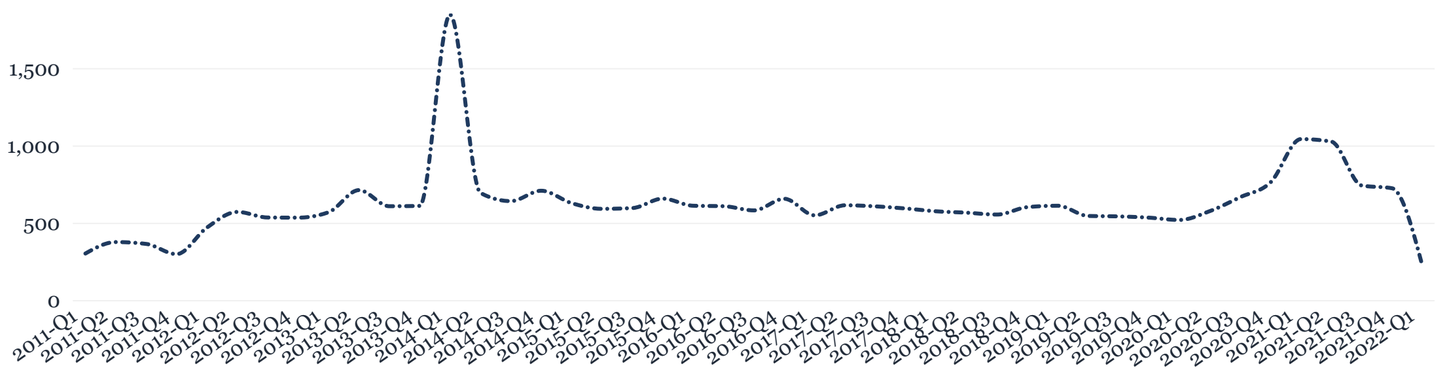
Encounters by Year



Encounters by Calendar Month(all years)



Quarterly Encounters



Year	Total Encounters	YoY Change %
2011	1,336	
2012	2,106	57.6%
2013	2,495	18.5%
2014	3,885	55.7%
2015	2,468	-36.5%
2016	2,452	-0.6%
2017	2,359	-3.8%
2018	2,293	-2.8%
2019	2,228	-2.8%
2020	2,519	13.1%
2021	3,530	40.1%
2022	220	

Footnote;2022 is partial(date ends 5 Feb 2022). February 2014 is an unusually high month (1,179 encounters);the data does not explain why.

Hospital Operations Encounter Classes Analysis

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Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

Prepared by Kukoyi Zainab Omotola

Total Encounters
27,891

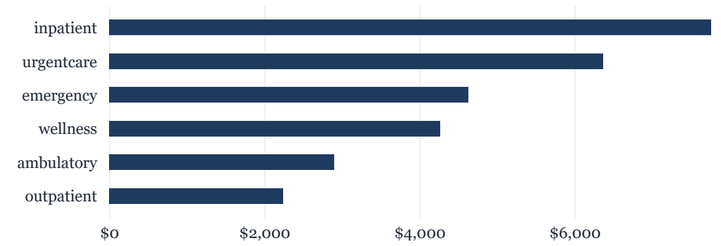
Total Claim Cost
$101.5M

Avg Claim Cost per Encounter
$3,640

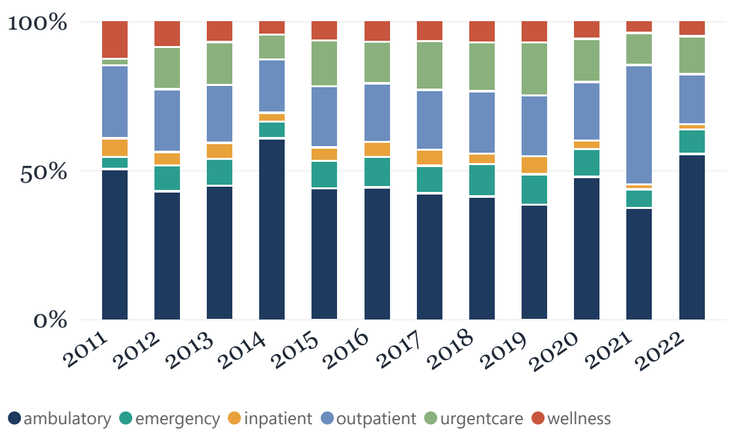
Urgent care is 13.1% of encounters but 23.0% of claim cost. Inpatient has the highest average claim cost ($7,761) and accounts for 58 of the 75 encounters over 24 hours.

Class	Encounters	Share %	Total Claim Cost	Cost Share %	Avg Cost	Encounters Over 24h	Over 24h %
ambulatory	12,537	44.9%	$36.28M	35.7%	$2,894	5	0.04%
emergency	2,322	8.3%	$10.75M	10.6%	$4,630	5	0.22%
inpatient	1,135	4.1%	$8.81M	8.7%	$7,761	58	5.11%
outpatient	6,300	22.6%	$14.09M	13.9%	$2,237	7	0.11%
urgentcare	3,666	13.1%	$23.35M	23.0%	$6,369	0	0.00%
wellness	1,931	6.9%	$8.23M	8.1%	$4,261	0	0.00%

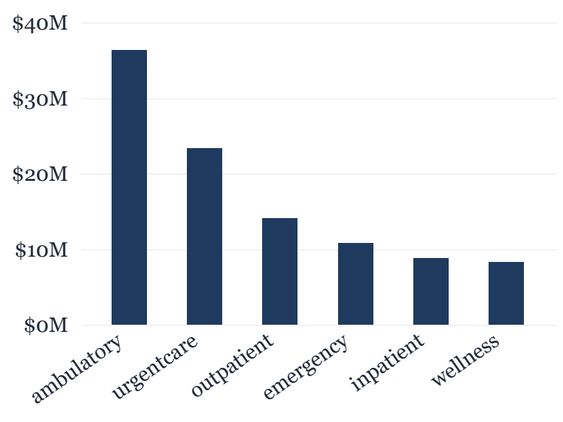
Claim Cost per Encounter



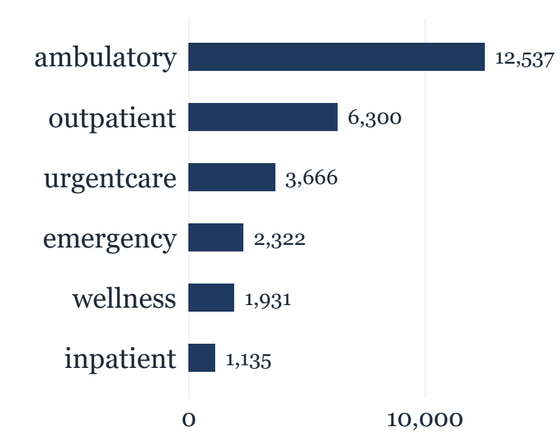
Class Mix by Year



Total Claim Cost by Class



Encounters by Class



Footnote;2022 is partial(data ends 5 Feb 2022). Class definitions are as recorded in the data

Hospital Operations Cost & Payer Coverage

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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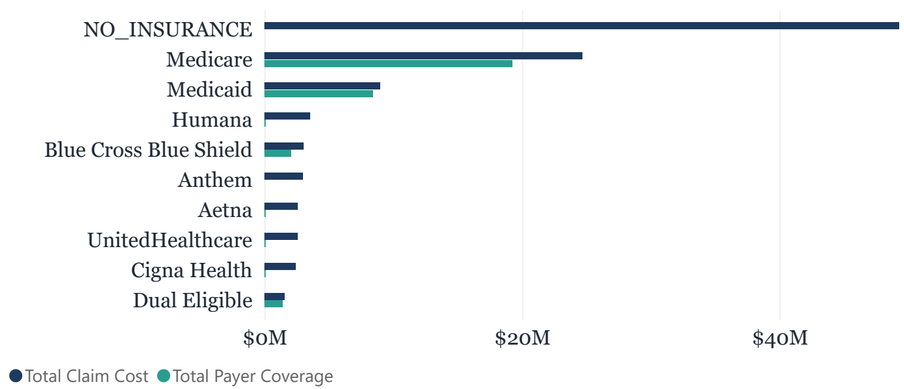
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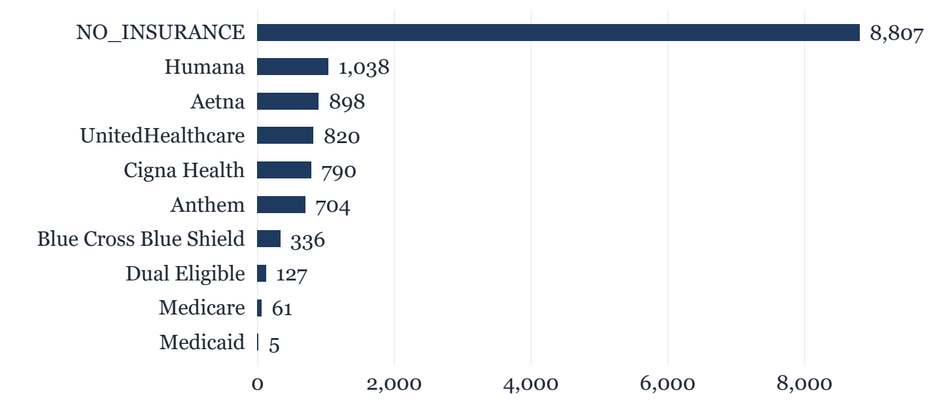
Prepared by Kukoyi Zainab Omotola

Total Payer Coverage	Payer Coverage %	Zero Coverage %	Cost Not Covered by Payer	Zero Coverage Encounters
$31.1M	30.6%	48.7%	$70.4M	13,586

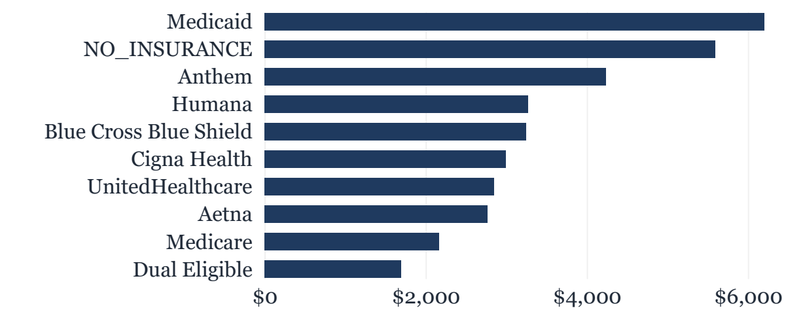
Claim Cost vs Payer Coverage by Payer



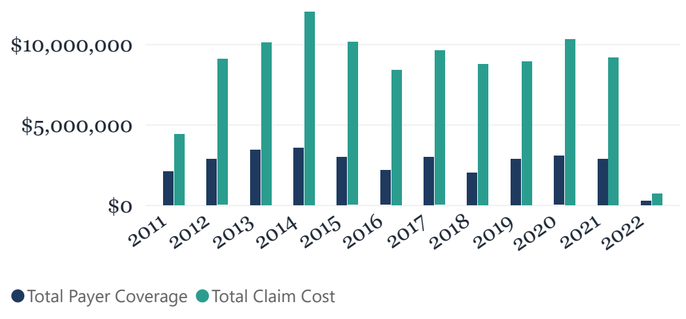
Zero Coverage Encounters by Payer



Avg Claim Cost per Encounter by Payer



Claim Cost vs Payer Coverage by Year



Key Takeaway

Five private insurers (Humana, Aetna, UnitedHealthcare, Cigna and Anthem) record 0.0% to 0.2% of claim cost as payer coverage, while Medicare, Medicaid and Dual Eligible record 78% to 94%. The data does not explain the difference.

Footnote;Zero payer coverage may reflect uninsured encounters, unrecorded coverage or unpaid claims; the data does not say which. 2022 is partial (data ends 5 Feb 2022).

Hospital Operations Procedures Utilisation

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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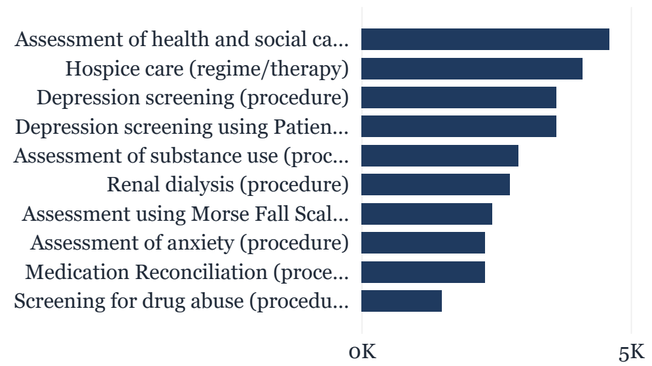
All

Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

Prepared by Kukoyi Zainab Omotola

Procedure Count	Avg Procedure Base Cost	Encounters with Procedure	Encounters without Procedure %
47,701	$2,212	14,670	47.4%

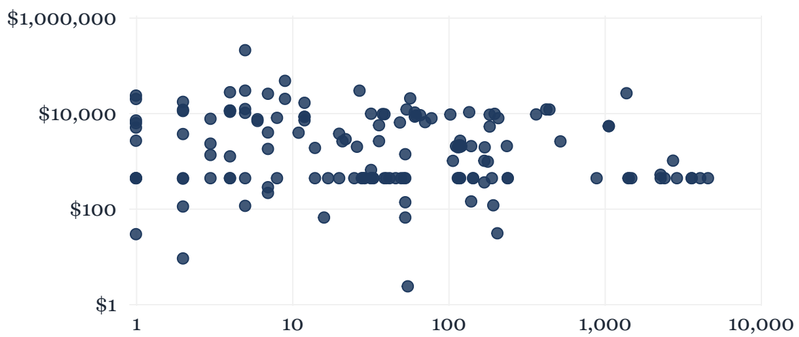
Top 10 Procedures by Frequency



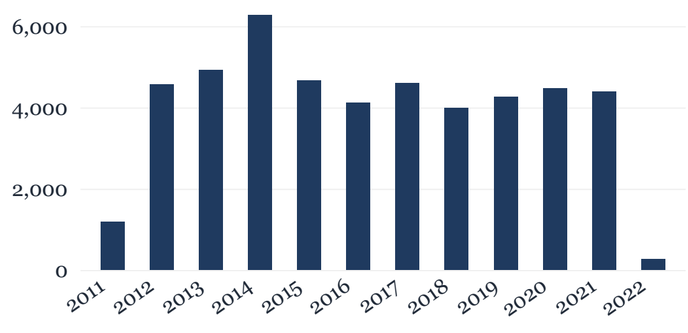
Highest Average Base Cost (min. 20 records)

DESCRIPTION	Procedure Count	Avg Procedure Base Cost	Total Procedure Base Cost
Hemodialysis (procedure)	27	$29,299.556	791,088.00
Electrical cardioversion	1,383	$25,903.111	35,824,002.00
Percutaneous mechanical thrombectomy of portal vein using fluoroscopic guidance	57	$20,228.035	1,152,998.00
Colonoscopy	424	$11,841.988	5,021,003.00
Subcutaneous immunotherapy	54	$11,835.500	639,117.00
Combined chemotherapy and radiation therapy (procedure)	444	$11,796.387	5,237,596.00
Cognitive and behavioral therapy (regime/therapy)	136	$10,388.081	1,412,779.00
Rectal polypectomy	61	$10,231.557	624,125.00
Induced termination of pregnancy	32	$9,641.000	308,512.00
Bone density scan (procedure)	198	$9,602.864	1,901,367.00

Frequency vs Average Base Cost



Procedures by Year



Key Takeaway

The ten most frequent procedures make up 63% of records, mostly routine assessments with a flat $431 base cost. Electrical cardioversion has the largest total base cost (1,383 records, average $25,903, about $35.8M). Base cost is not claim cost.

Footnote;Procedure base cost is not the same as claim cost. 47.4% of encounters have no procedure record. Averages use procedures with at least 20 records. 2022 is partial (data ends 5 Feb 2022).

Hospital Operations Data Notes & KPI Definitions

Encounters, costs, payer coverage and procedures | Jan 2011 to Feb 2022

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Data: Jan 2011 to Feb 2022, one hospital. 2022 is a partial year.

Prepared by Kukoyi Zainab Omotola

KPI name	Definition	Calculation
Total Encounters	Number of encounter records.	Count of rows in encounters.
Procedure Count	Number of procedure records.	Count of rows in procedures.
Over 24h %	Encounters where STOP minus START is more than 24 hours, divided by all encounters.	Encounters Over 24h / Total Encounters. Duration = (STOP - START) in hours.
Avg Procedure Base Cost	Average of procedure base cost.	Sum of BASE_COST / number of procedures.
Total Claim Cost	Sum of total claim cost across encounters.	Sum of TOTAL_CLAIM_COST.
Avg Claim Cost per Encounter	Total claim cost divided by number of encounters.	Total Claim Cost / Total Encounters.
Payer Coverage %	Total payer coverage divided by total claim cost.	Total Payer Coverage / Total Claim Cost.
Zero Coverage %	Encounters with PAYER_COVERAGE equal to 0, divided by all encounters.	Zero Coverage Encounters / Total Encounters.

Data Quality Summary

- Source: six CSV files from AnalystLab Africa: 27,891 encounters, 974 patients, 47,701 procedures, 10 payers, 1 organization
- Period: 2 Jan 2011 to 5 Feb 2022
- No duplicate records and no orphan keys
- All start and stop times valid; none stops before it starts
- Reason fields blank in 70% of encounters and 77% of procedures; not used
- 89 encounters have a claim cost of zero; coverage never exceeds claim cost
- 48.7% of encounters have zero recorded payer coverage
- 47.4% of encounters have no procedure record
- 51% of procedure start times fall outside their encounter's time window, so procedures are placed by encounter date
- Four procedure codes appear under more than one description
- No rows removed and no values replaced

Limitations

- One hospital only; 2022 is partial (data ends 5 Feb)
- February 2014 has 1,179 encounters against about 200 in a typical month; the data does not explain it
- Zero payer coverage may mean uninsured, unrecorded or unpaid
- Procedure base cost is not claim cost and should not be added to it
- The median encounter lasts 15 minutes; of the 75 over 24 hours, 13 last over 30 days, so treat "over 24 hours" with care
- Currency is not stated
- Records appear synthetic or anonymised
- No clinical outcomes, so no clinical claims

Method & Handling

- Tools: Power Query (loading and cleaning), DAX (measures), Excel and SQL (validation)
- Timestamps are UTC; duration = STOP minus START, in hours
- Encounters link to patients, payers, organisations and dates; procedures link to encounters
- Personal columns were removed from patients before loading
- Key totals were checked independently and filters tested
- Prepared by Kukoyi Zainab Omotola; AI assistance (Claude) was used for planning, profiling and documentation